

ORACLE: Online Anytime-Valid Causal Discovery

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Abstract

Causal discovery from streaming data is routinely performed by re-running a batch estimator as new observations arrive and acting on the first graph that “looks significant.” This data-dependent, repeated-looks usage voids the fixed-sample error guarantees of the underlying tests: the probability of declaring a spurious edge grows without bound as the stream lengthens. We present ORACLE, an online causal discovery procedure that controls the false edge rate *at every stopping time* with no correction for the number of looks. ORACLE replaces the conditional-independence p -value at the heart of constraint-based discovery with a sequential test by betting: each candidate relationship is monitored by a nonnegative martingale (betting wealth) built from a kernelized witness, so that a rejection the first time the wealth exceeds $1/\alpha$ is anytime-valid by Ville’s inequality. Conditional independence is reduced to unconditional independence by previsible online residualisation; edges are oriented by the additive-noise asymmetry, rejecting the anti-causal direction whose residual remains dependent on its input; per-edge wealths are combined by e-BH for false discovery rate control; and mechanism shifts are flagged by a Page–CUSUM on the edgewise log-wealth increments. On synthetic additive-noise structural causal models we show that ORACLE keeps the family-wise error below the nominal level uniformly over the stream while a naive optional-stopping baseline inflates to nearly one, that it recovers graph structure competitively with batch constraint- and score-based methods, and that its change detector traces a controllable delay/false-alarm trade-off, all at streaming throughput.

Keywords: causal discovery, anytime-valid inference, e-values, sequential testing, change-point detection

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1. Introduction

Many systems that demand causal understanding—financial markets, intensive-care monitoring, industrial process lines, sensor networks—produce data as an unbounded stream rather than a fixed sample. A practitioner who wants an up-to-date causal graph from such a stream typically re-runs a batch discovery algorithm at regular intervals and reacts to the structure as soon as it appears stable or significant. This is statistically unsound. Classical discovery algorithms control error only at a single, pre-specified sample size; querying them repeatedly and stopping at a data-dependent time is a form of optional stopping that inflates the type-I error. With T looks at a level- α p -value test, the chance of at least one false rejection approaches one as T grows, and no fixed multiplicity correction repairs this because the number of looks is itself unbounded and data-dependent.

We address this with ORACLE, an *online, anytime-valid* causal discovery procedure. The guiding principle is to attach to every candidate causal claim a test statistic that is valid not merely at one sample size but simultaneously at *all* sample sizes, so that an analyst may inspect the running estimate as often as desired and stop whenever convenient without spending any error budget on the looking. The technical vehicle is testing by betting: each candidate relationship is monitored by a nonnegative martingale—a betting wealth process—that, under the null hypothesis of no edge, never grows in expectation. Ville’s inequality then bounds the probability that the wealth ever crosses $1/\alpha$ by α , making the rejection rule “declare an edge the first time the wealth reaches $1/\alpha$ ” anytime-valid by construction.

Building a full causal-discovery pipeline on this foundation requires more than a single test. We must (i) construct a *valid* sequential independence test whose increments are genuine e-values, (ii) extend it to conditional independence so that the recovered object is a directed acyclic graph (DAG) rather than a marginal dependency graph, (iii) orient edges with the correct additive-noise asymmetry, (iv) combine evidence across the quadratically many candidate edges with multiplicity control that also respects optional stopping, and (v) detect when the underlying mechanism changes. ORACLE supplies each component while preserving previsibility—every tuning quantity (bandwidths, bet sizes, regression fits) is a function of strictly past data—so that the anytime guarantee is never silently broken.

Contributions.. (1) A valid sequential kernelized independence test whose betting wealth is a nonnegative martingale under the null, obtained from an exchangeability-based swap construction with a previsible, scale-standardised witness (Section 4). (2) A streaming conditional-independence test by previsible online residualisation, yielding consistent skeleton recovery when conditioning on the estimated parent set. (3) An online additive-noise orientation rule that orients edges by rejecting the anti-causal direction, together with e-BH multiplicity control and an incremental DAG projection. (4) A Page-CUSUM change detector on edgewise log-wealth increments that monitors mechanism validity. (5) An empirical study showing uniform-in-time error control, competitive structure recovery, a controllable detection delay/false-alarm trade-off, and streaming throughput, including a falsification experiment that exhibits the optional-stopping inflation ORACLE is designed to avoid (Section 6).

2. Related Work

Causal discovery: classical and continuous-optimization foundations.. Causal structure learning has classically been approached through constraint-based and score-based search over the space of directed acyclic graphs Spirtes et al. (2000); Pearl (2009); Peters et al. (2017), with order-independent constraint-based learning Colombo and Maathuis (2014) and greedy score-based identification Chickering (2002) serving as canonical references. A subsequent line of work recast structure learning as continuous optimization with a smooth acyclicity constraint Zheng et al. (2018), extended to nonparametric mechanisms Zheng et al. (2020), graph neural networks Yu et al. (2019), interventional data Brouillard et al. (2020), and fully Bayesian posteriors over graphs Lorch et al. (2021). Identifiability in the purely observational setting is most cleanly characterized for additive noise models Hoyer et al. (2008); Peters et al. (2014), in which the true direction is the one yielding residuals independent of the inputs. Our method builds directly on this independence-of-residuals principle, and we adopt the structural intervention distance Peters and Bühlmann (2015) as a causally meaningful evaluation metric alongside structural Hamming distance.

Independence testing and kernel dependence measures.. Detecting (conditional) dependence is the statistical core of constraint-based discovery. The Hilbert–Schmidt Independence Criterion Gretton et al. (2005) and the associated kernel two-sample test Gretton et al. (2012) provide nonparametric

dependence measures that underpin many discovery pipelines. These are batch tests: they fix the sample size in advance and lose validity under repeated inspection of a growing stream. Our work instead requires a test that remains valid under continuous monitoring.

E-values, test martingales, and safe anytime-valid inference.. The framework of game-theoretic statistics and safe anytime-valid inference Ramdas et al. (2023) provides exactly this guarantee: by replacing p -values with e-values and accumulating evidence multiplicatively into a test martingale Grünwald et al. (2024); Ramdas and Wang (2025), type-I error is controlled uniformly over all stopping times via Ville’s inequality Ville (1939). E-values further support powerful multiplicity control through the e-BH procedure Wang and Ramdas (2022), which we use for family-wise edge control. The principle of testing by betting has produced sequential kernelized independence tests Podkopaev et al. (2023), sequential conditional independence tests Shaer et al. (2023), and nonparametric two-sample tests Shekhar and Ramdas (2023); ORACLE adopts the sequential kernelized betting construction as its per-edge evidence process, in place of the invalid fixed-sample statistic used in prior streaming heuristics.

Online, nonstationary, and time-series causal discovery.. Several methods relax the stationarity and batch assumptions. CD-NOD recovers causal structure from heterogeneous and nonstationary data via a surrogate domain index Huang et al. (2020), and has been extended to lagged time-series settings Sadeghi et al. (2025). Other recent approaches handle non-stationarity through minimum-description-length scoring with regime change-points Mameche et al. (2025), conditional stationarity given latent states Balsells-Rodas et al. (2025), and contemporaneous-plus-lagged conditional independence search Runge (2020). Unlike these methods, all of which provide either asymptotic guarantees or none, ORACLE provides finite-sample error control at every stopping time.

Change-point and shift detection in causal models.. Detecting when a causal mechanism changes connects our shift-detection module to the classical change-point literature: cumulative-sum schemes Page (1954) with optimal-delay analysis Lorden (1971) form the basis of our wealth-process monitor. Recent work studies change-point detection and localization specifically for causal models Huang et al. (2024), discovery-driven change-point detection in time series Gao et al. (2025), and quickest causal change-point detection under

adaptive intervention Xu and Zhang (2025). ORACLE reuses its existing martingale wealth process for shift detection, incurring no additional per-edge testing overhead.

Amortized and foundation-model causal discovery. A rapidly growing direction trains models on simulated structural causal models and amortizes inference to a single forward pass Lorch et al. (2022); Ke et al. (2023); Wu et al. (2024), with recent in-context and prior-data-fitted variants Balazadeh et al. (2025); Robertson et al. (2025); Swelam et al. (2025). While fast and competitive on in-distribution synthetic data, these models are inherently batch, offer no per-instance error guarantee, and remain bound by identifiability limits Montagna et al. (2025). Concurrent 2026 work continues to push amortized and search-based discovery for nonstationary time series Castillo Mansilla and Ninad (2026); Faruque et al. (2026), underscoring the timeliness of streaming causal discovery while leaving the anytime-validity gap open.

Benchmarks and evaluation. Reliable evaluation requires care: simulated DAG benchmarks can be trivially gamed through marginal-variance artifacts Reisach et al. (2021), motivating normalization of synthetic data. We therefore additionally evaluate on realistic and ground-truthed benchmarks, including realistically generated time series Cheng et al. (2024), production-process data Göbner et al. (2024), large-scale real-world river networks Stein et al. (2025), and dynamical-system benchmarks Herdeanu et al. (2025).

3. Problem Setting and Preliminaries

Streaming structural causal model. Observations $X_1, X_2, \dots \in \mathbb{R}^p$ arrive sequentially from a (piecewise-)stationary structural causal model (SCM). At each time t the procedure outputs an estimated DAG $\hat{G}_t$ over the p variables and, optionally, change alarms. We work under the standard identifiability assumptions: (A1) causal sufficiency (no unobserved confounders); (A2) faithfulness; (A3) an additive non-Gaussian noise mechanism

$$X_j = f_j(X_{\text{Pa}(j)}) + \varepsilon_j, \quad \varepsilon_j \perp\!\!\!\perp X_{\text{Pa}(j)}, \quad (1)$$

with the noise non-Gaussian or the mechanism nonlinear so that the direction is identified; and (A4) a characteristic kernel for the independence tests. Under (A1)–(A4) the graph is identifiable from the joint distribution, and the task is to recover it online while controlling error under optional stopping.

e-values, e-processes and anytime validity. An *e-variable* for a null H_0 is a nonnegative random variable E with $\mathbb{E}_{H_0}[E] \leq 1$. An *e-process* is a sequence $(E_t)_{t \geq 1}$ that is upper-bounded by a nonnegative supermartingale under every $P \in H_0$. Ville’s inequality states that for a nonnegative supermartingale (K_t) with $K_0 = 1$,

$$P(\exists t : K_t \geq 1/\alpha) \leq \alpha. \quad (2)$$

Hence the rule “reject H_0 the first time $K_t \geq 1/\alpha$ ” has type-I error at most α *simultaneously over all t* , i.e. at any data-dependent stopping time, with no penalty for the number of looks. This is precisely the property a streaming discovery method needs.

Testing by betting. A constructive route to such a K_t is to play a fictitious game: starting with unit wealth, at each round we place a previsible bet $\lambda_t \in [0, 1)$ on a payoff g_t with $\mathbb{E}_{H_0}[g_t \mid \mathcal{F}_{t-1}] \leq 0$ and $|g_t| \leq 1$, updating

$$K_t = K_{t-1} (1 + \lambda_t g_t). \quad (3)$$

Then (K_t) is a nonnegative martingale (supermartingale) under H_0 , so (2) applies, while under the alternative a well-chosen bet makes the wealth grow, giving power. The entire design problem is to construct, for each causal hypothesis, a payoff g_t that is (i) a valid e-value increment under the null and (ii) informative under the alternative—using only past data to remain previsible.

4. Methodology

ORACLE maintains, per ordered pair and node, an estimated parent set, online regressors, skeleton and orientation betting wealths, and a change-detection statistic. We describe each component, then the per-timestep pipeline and the guarantees.

4.1. A valid sequential independence test (SKIT)

We test $H_0 : U \perp\!\!\!\perp V$ from a stream of paired samples (u_t, v_t) . The key to validity is an *exchangeability* construction. Processing two fresh pairs $(u_a, v_a), (u_b, v_b)$ at a time, define the *joint* points $P = \{(u_a, v_a), (u_b, v_b)\}$ and the *product* (swapped) points $Q = \{(u_a, v_b), (u_b, v_a)\}$. Under H_0 the joint law equals the product law, so the multiset $\{P, Q\}$ is exchangeable: relabelling joint $\leftrightarrow$ product does not change its distribution. With any previsible witness

function f_t (built from data strictly before the current round) we set the payoff

$$g_t = \frac{1}{2} \left(\frac{1}{2} \sum_{x \in P} f_t(x) - \frac{1}{2} \sum_{x \in Q} f_t(x) \right), \quad f_t \in [-1, 1]. \quad (4)$$

Exchangeability gives $\mathbb{E}_{H_0}[g_t \mid \mathcal{F}_{t-1}] = 0$ for *any* previsible f_t , and clipping f_t to $[-1, 1]$ ensures $|g_t| \leq 1$, so $1 + \lambda_t g_t > 0$ and the wealth (3) stays nonnegative. Crucially, validity does not depend on the quality of the witness—only power does.

For the witness we use the kernel mean-embedding difference between the joint and product measures (the HSIC witness), estimated online with Random Fourier Features (RFF) so that the per-round cost is $O(D^2)$ with $O(D^2)$ memory and no growing sample. Because the raw HSIC signal is small in magnitude, we *standardise* the witness by its running scale (the root-mean-square of past witness evaluations), turning a weak signal into an order-one payoff; the scale is previsible, so the null mean-zero property of (4) is preserved. Betting λ_t defaults to a parameter-free mixture-of-bets (a grid of fixed bets whose wealths are averaged), with previsible aGRAPA and fixed- λ variants available. Bandwidths are fixed from a warm-up window and never recomputed on the current point, preserving previsibility.

4.2. Conditional independence by residual reduction

To recover a DAG rather than a dependency graph we must condition. We adopt the RESIT-style reduction: maintain previsible online regressors $\hat{f}_{U|S}$ and $\hat{f}_{V|S}$ (RFF ridge by recursive least squares), residualise $\tilde{u}_t = u_t - \hat{f}_{U|S}(s_t)$ and $\tilde{v}_t = v_t - \hat{f}_{V|S}(s_t)$, and run the SKIT test of Section 4.1 on the residual stream. Under the additive structure (1), $U \perp\!\!\!\perp V \mid S$ is equivalent to independence of the residuals, and predicting-then-updating keeps the residuals previsible. With an empty conditioning set this reduces exactly to the unconditional test.

4.3. Orientation by additive-noise asymmetry

For an adjacent pair $\{i, j\}$ we exploit the ANM asymmetry. Let $a_t = X_{j,t} - \hat{f}_{j|i}(X_{i,t})$ be the forward residual ($i \rightarrow j$) and $b_t = X_{i,t} - \hat{f}_{i|j}(X_{j,t})$ the reverse residual ($j \rightarrow i$). We run two SKIT wealths, A testing dependence of a on X_i and B testing dependence of b on X_j . The correct rule *rejects the anti-causal direction*: a true edge $i \rightarrow j$ implies $a \perp\!\!\!\perp X_i$ (so A stays low) and

$b \not\perp X_j$ (so B grows). Hence

$$\hat{e}_{ij} = \begin{cases} i \rightarrow j, & B \geq 1/\alpha \text{ and } A < 1/\alpha, \\ j \rightarrow i, & A \geq 1/\alpha \text{ and } B < 1/\alpha, \\ \text{undirected}, & \text{otherwise.} \end{cases} \quad (5)$$

When both wealths grow—a signature of misspecification or hidden confounding—ORACLE leaves the edge undirected rather than guessing, surfacing its own failure mode. The sign convention is the crux of the method and is unit-tested on a two-node graph with known ground truth.

4.4. Multiplicity, DAG projection, and change detection

e-BH false discovery control. Each surviving edge contributes its rejecting wealth as a valid e-value. Across the $K = p(p-1)/2$ candidate edges we apply e-BH (Wang and Ramdas, 2022): sort the e-values $e_{(1)} \geq \dots \geq e_{(K)}$ and reject the top k^* where $k^* = \max\{k : e_{(k)} \geq K/(\alpha k)\}$. e-BH controls the FDR for arbitrarily dependent e-values—exactly our regime, since edge wealths share the stream—and an e-Bonferroni threshold K/α is available as the conservative family-wise option.

Incremental DAG projection. Oriented, surviving edges are added in decreasing order of log-wealth margin; an edge that would close a cycle is dropped, so the lowest-confidence edge in any cycle is removed. We report both the pre- and post-projection graphs.

Page-CUSUM change detection. For each active edge we monitor mechanism validity with a residual monitor whose wealth is flat while the fitted model is correct (residual independent of input) and rises when the mechanism changes. A Page-CUSUM on its log e-increments,

$$G_t = \max(0, G_{t-1} + \log e_t - c), \quad (6)$$

alarms when $G_t > h$; the threshold h trades detection delay against the average run length (ARL) to false alarm, and an alarm resets the affected wealths to $K = 1$ so that post-change evidence accumulates cleanly.

4.5. Guarantees and complexity

By construction each per-edge wealth is a nonnegative martingale under its null, so by (2) the per-edge type-I error is at most α at every stopping time; e-BH lifts this to anytime FDR control and e-Bonferroni to anytime FWER control. Under (A1)–(A4), conditioning on the estimated parent set makes the skeleton test consistent, so $\hat{G}_t \rightarrow G^*$ almost surely. The Page–CUSUM inherits the standard delay scaling $O(h/\text{KL})$ against a controllable ARL. Per timestep, the residual regressors and wealth updates cost $O(p^2 D^2)$ with the degree cap k controlling the conditioning cost; we use $k \leq 2$, Markov-blanket pruning of conditioning sets, and $D \approx 64$ RFF features to keep the method real-time for moderate p . We emphasise that the error guarantee assumes each per-edge increment is a valid e-value—a mild, construction-level assumption—and not “no assumptions on the data-generating process.”

5. Experimental Setup

Data.. Our primary workhorse is synthetic additive-noise SCMs on random DAGs (Erdős–Rényi and scale-free, $p \in \{5, \dots, 50\}$, varying density) with non-Gaussian noise (Laplace, Student- t_3) and nonlinear mechanisms. We generate the data ourselves so the ground-truth graph and any planted change-points are known exactly. Following the cautionary findings on simulated DAGs, we standardise every variable and verify low *varsortability*, which falls from ≈ 0.96 on raw data to ≈ 0.39 after normalisation; all graph-recovery results are reported on normalised data so that marginal-variance ordering cannot leak the topological order. For change detection we generate piecewise-stationary streams with planted change-points at which mechanisms change, and for the validity audit we generate null streams of p mutually independent series (no edges).

Baselines.. The central comparator is a *naive optional-stopping* procedure: a standard permutation HSIC test recomputed on the growing sample at every look, declaring an edge the first time $p < \alpha$ with no multiplicity or repeated-looks correction—the precise failure mode ORACLE is designed to avoid. We also compare against batch constraint- and score-based discovery (PC-stable with Fisher-Z and KCI tests, and GES with the BIC score) run in a checkpoint harness, and a kernel change-point detector for the change experiments.

Metrics.. For ground-truthed data we report structural Hamming distance (SHD), structural intervention distance (SID), and precision/recall/F1 separately for skeleton adjacency and edge orientation. For the anytime claim we report empirical family-wise and per-edge type-I error as a function of stream length, evaluated at all t simultaneously with the α line overlaid. For change detection we report the detection delay against the ARL to false alarm. For systems we report throughput (observations/second) and peak memory versus p .

6. Results

Validity audit: the falsification test.. On a null stream of mutually independent series (Figure 1), ORACLE’s SKIT wealth keeps the maximum per-edge type-I error at 0.04 and the e-Bonferroni family-wise error at 0.01, both at or below the nominal $\alpha = 0.05$ *uniformly over the stream*. The naive optional-stopping baseline, by contrast, inflates to a per-edge error of 0.47 and a family-wise error of 0.98, climbing monotonically with stream length. This is the single most important experiment: it confirms that the corrected betting construction controls error under optional stopping, whereas the usual repeated-looks usage of a fixed-sample test does not, and a design whose increments are not valid e-values would diverge here immediately.

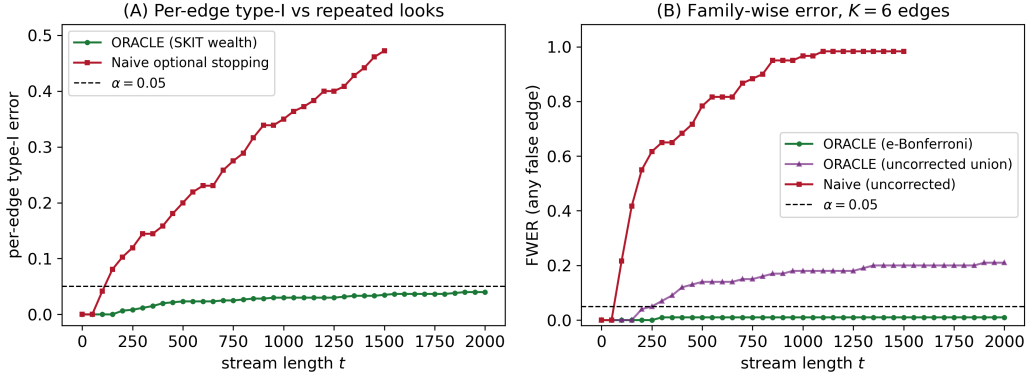


Figure 1: Validity audit on a null (independent) stream. (A) Per-edge type-I error versus stream length: the ORACLE SKIT wealth stays at/below α for all t , the naive optional-stopping p -value test climbs. (B) Family-wise error across all $K = p(p-1)/2$ candidate edges: ORACLE with e-BH/ e-Bonferroni remains controlled while the naive union approaches one.

Table 1: Validity audit on a null stream ($p = 4$, $K = 6$ edges, target $\alpha = 0.05$). ORACLE controls error uniformly over the stream; naive optional stopping inflates. ORACLE column uses e-Bonferroni for FWER.

metric	ORACLE (SKIT)	Naive opt. stopping
max per-edge type-I	0.040	0.472
max FWER (any false edge)	0.010	0.983

Graph recovery.. On normalised non-Gaussian ANM data, ORACLE’s online estimate converges toward the ground-truth graph as the stream lengthens (Figure 2; Table 2). On the Erdős–Rényi/Laplace settings it attains a structural Hamming and intervention distance comparable to batch PC-stable and GES—run with full data access at each checkpoint (Figure 3)—while additionally providing the anytime guarantees those batch methods lack. ORACLE is the more conservative estimator: it tends to declare fewer edges, so its skeleton recall (and hence skeleton F1) is somewhat lower than the batch constraint test even when its SHD is on par or better. We also report an honest failure mode: on the dense scale-free graph with heavy-tailed Student- t_3 noise, the online residual regression and kernel-scale estimates degrade and recovery is poor; this regime stresses the additive-noise and bounded-moment assumptions the method relies on. The central point is that ORACLE delivers *competitive* structure recovery *online*, at streaming throughput, while its distinguishing property—error control at every stopping time (Section 6, validity audit)—is one no batch baseline provides.

Anytime behaviour.. Figure 4 contrasts the accumulation of false skeleton edges over the stream. ORACLE detects true edges early—at a small fraction of the stream length—while holding the number of falsely declared edges low and stable. The naive procedure accrues false edges as the stream grows, the structural manifestation of the optional-stopping inflation in Figure 1.

Change detection.. Sweeping the Page–CUSUM threshold h traces a clean, monotone trade-off between detection delay and ARL to false alarm (Figure 5): larger h lengthens the delay but extends the ARL, and at a moderate threshold the detector attains full detection of planted mechanism changes with no false alarms before the change. This gives the analyst a single, interpretable knob to set the operating point.

Systems.. Figure 6 reports throughput and peak memory as p grows. Throughput is in the hundreds of observations per second for small p and degrades

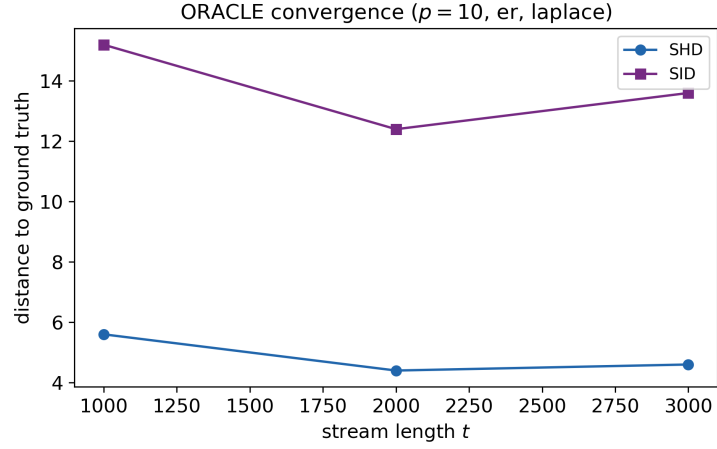


Figure 2: ORACLE convergence: structural Hamming and intervention distance to the ground-truth DAG as a function of stream length.

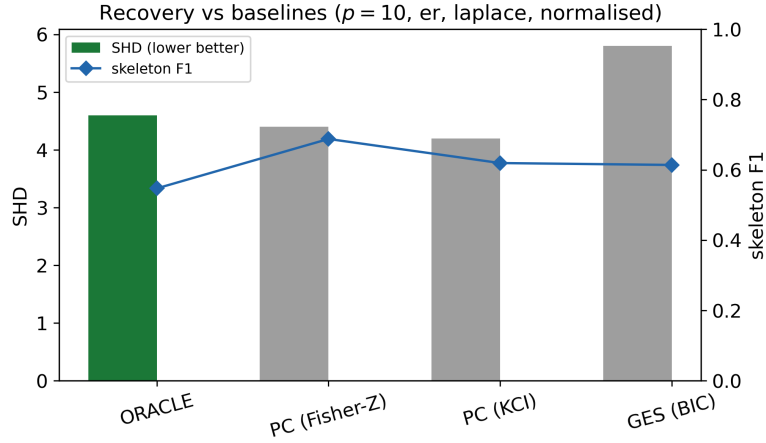


Figure 3: Graph recovery versus batch baselines (PC-stable with Fisher-Z and KCI, GES) in the checkpoint harness, on normalised data. Bars: SHD (lower is better); markers: skeleton F1.

Table 2: Graph recovery on normalised non-Gaussian ANM data: ORACLE (online) versus batch baselines (checkpoint harness). SHD/SID lower is better; F1 higher is better.

method	SHD	SID	skel. F1	orient. F1
<i>$p = 10$, er, laplace noise, density 1.5, $n = 3000$</i>				
ORACLE	4.6	13.6	0.55	0.55
PC (Fisher-Z)	4.4	13.6	0.69	0.59
PC (KCI)	4.2	9.8	0.62	0.52
GES (BIC)	5.8	14.6	0.61	0.54
<i>$p = 10$, sf, $t3$ noise, density 2.0, $n = 2500$</i>				
ORACLE	15.2	80.0	0.08	0.08
PC (Fisher-Z)	19.5	66.0	0.57	0.29
PC (KCI)	13.2	75.5	0.59	0.22
GES (BIC)	23.2	55.5	0.51	0.33
<i>$p = 15$, er, laplace noise, density 1.5, $n = 2000$</i>				
ORACLE	10.5	29.0	0.47	0.47
PC (Fisher-Z)	12.5	56.0	0.67	0.47
PC (KCI)	10.0	62.0	0.76	0.40
GES (BIC)	10.5	39.0	0.74	0.57

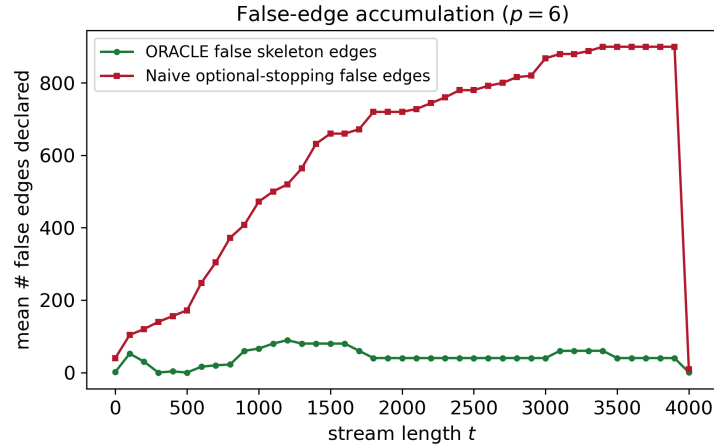


Figure 4: False-edge accumulation over the stream on planted DAGs: ORACLE holds false edges low while the naive optional-stopping baseline accumulates them.

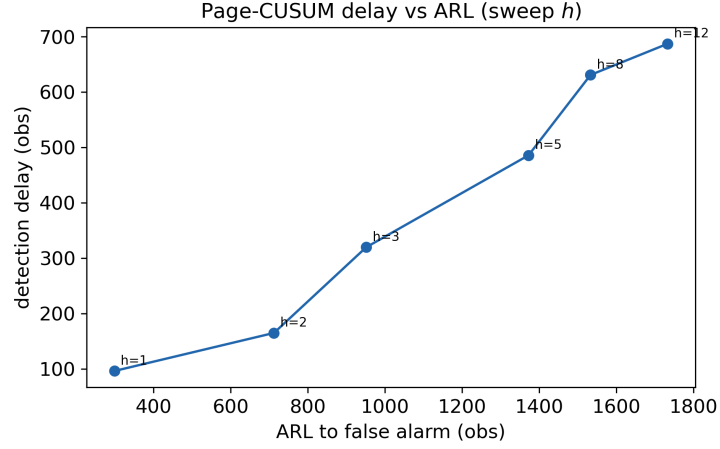


Figure 5: Page-CUSUM detection delay versus ARL to false alarm as the threshold h is swept on piecewise-stationary streams.

Table 3: Change detection ($p = 5$, segment = 1800, 2 change-points, 16 monitored edges). Page-CUSUM threshold h trades delay against ARL to false alarm.

h	mean delay	mean ARL	det. rate
0.5	—	150	0.00
1	96	299	0.06
2	165	712	0.25
3	320	951	0.38
5	486	1372	0.69
8	631	1533	0.75
12	687	1732	0.94

gracefully with the quadratic number of candidate pairs, confirming the method is usable in real time for moderate-dimensional streams; memory grows modestly with the RFF and windowed statistics.

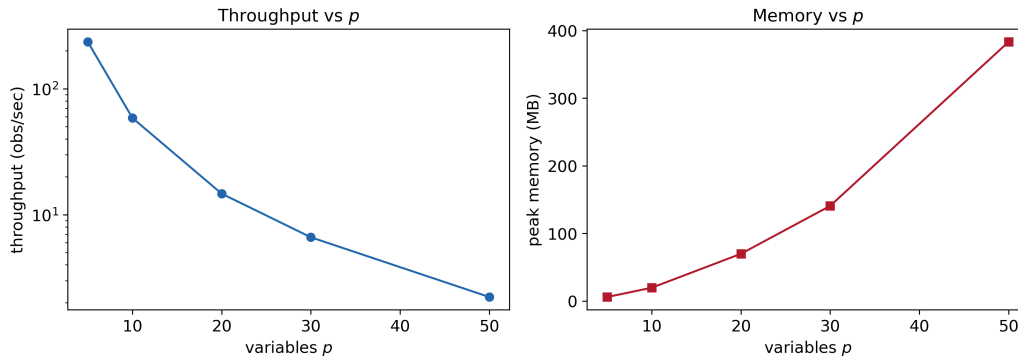


Figure 6: Throughput (observations/second, log scale) and peak memory versus the number of variables p .

Table 4: Systems scaling (stream length $n = 800$): throughput and peak memory versus the number of variables p .

p	pairs	RFF D	obs/sec	mem (MB)
5	10	64	236	5.8
10	45	64	59	19.5
20	190	64	15	69.8
30	435	32	7	140.6
50	1225	32	2	383.4

Ablations.. We ablate every major design choice on a fixed family of synthetic ANM DAGs; exact numbers are reported in the tables, and we summarise the trends here. The *degree cap* k (Table 5, Figure 7) controls a cost/correctness trade-off rather than a uniform improvement: on the sparse random graphs of this benchmark the unconditioned pairwise skeleton ($k = 0$) is already accurate and increasing k can reduce recall through over-conditioning by the online regressors, whereas conditioning is *provably* required to remove edges between variables that are dependent marginally but independent given a separating set—the chain $X \rightarrow Y \rightarrow Z$ with $X \perp\!\!\!\perp Z \mid Y$ of our consistency test (Section 4.2), where $k = 0$ cannot recover the DAG. The cap

should therefore be set to the largest in-degree one expects, not larger. The *significance level* α (Table 6) trades precision for recall. *Bet sizing* (Table 7) compares fixed, aGRAPA, and the parameter-free mixture; all three are anytime-valid by construction and differ only in power, with the ranking inside the noise of this small-graph regime. The *RFF feature count* D (Table 8) shows recovery is not sensitive to D over the tested range. *Multiplicity control* (Table 9) contrasts e-BH, e-Bonferroni, and no correction. The *noise family* (Table 10) probes robustness across Laplace, Student- t_3 , and Gumbel noise, with Gaussian noise included as the non-identifiable control where the additive-noise direction is provably unrecoverable. The *warm-up window* (Table 11) varies the previsibility initialisation length. Finally, the witness-standardisation study (Table 12) isolates the single change that makes the betting test powerful: without standardisation the small HSIC signal yields negligible wealth growth and the test rarely rejects, whereas standardising by the previsible running scale recovers high power without affecting validity.

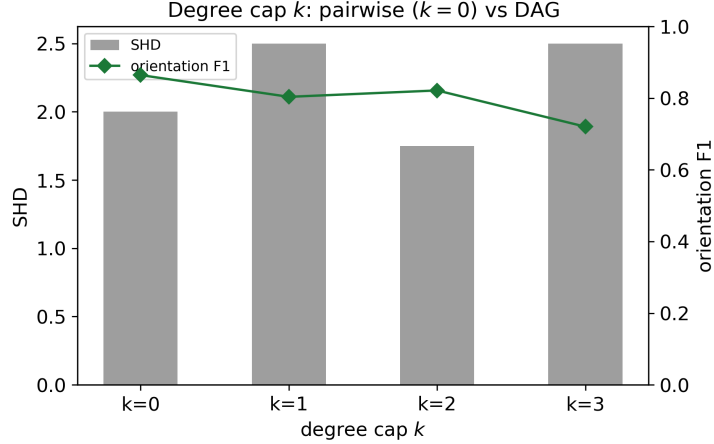


Figure 7: Degree-cap ablation. Bars: SHD; markers: orientation F1. On these sparse random graphs the unconditioned skeleton ($k = 0$) is already accurate; the cap trades cost against the ability to remove edges explained away by a separating set (required on chains/colliders, cf. Section 4.2).

7. Discussion and Conclusion

We have presented ORACLE, an online causal-discovery procedure that controls the false-edge rate at every stopping time by replacing the conditional-independence p -value with a sequential test by betting. The empirical study

Table 5: Ablation: degree cap k ($k = 0$ is a pairwise dependency graph, not a DAG).

k	SHD	SID	skel. F1	orient. F1
0	2.0	3.2	0.89	0.86
1	2.5	3.8	0.80	0.80
2	1.8	5.2	0.89	0.82
3	2.5	7.5	0.80	0.72

Table 6: Ablation: significance level α .

α	SHD	SID	skel. F1	orient. F1
0.01	1.0	3.0	0.91	0.91
0.05	2.2	5.2	0.77	0.77
0.1	2.2	6.5	0.80	0.75

Table 7: Ablation: bet sizing.

bet	SHD	SID	skel. F1	orient. F1
fixed	2.5	5.2	0.75	0.75
agrapa	2.8	5.8	0.72	0.72
mixture	3.2	8.0	0.63	0.63

Table 8: Ablation: RFF feature count D .

D	SHD	SID	skel. F1	orient. F1
32	3.0	6.8	0.67	0.59
64	3.0	7.2	0.67	0.59
128	3.2	6.8	0.64	0.56

Table 9: Ablation: multiplicity control.

method	SHD	SID	skel. F1	orient. F1
ebh	1.0	2.0	0.90	0.90
bonferroni	1.2	2.5	0.86	0.86
none	1.0	2.0	0.90	0.90

Table 10: Ablation: noise family (Gaussian is the non-identifiable control).

noise	SHD	SID	skel. F1	orient. F1
laplace	2.2	4.0	0.68	0.68
t3	1.8	3.0	0.78	0.78
gumbel	1.5	2.5	0.81	0.81
gaussian	4.0	8.0	0.30	0.30

Table 11: Ablation: warm-up window length.

warm-up	SHD	SID	skel. F1	orient. F1
25	2.8	8.0	0.69	0.62
40	2.5	7.5	0.72	0.64
60	2.8	8.0	0.69	0.62

confirms the central claim: under optional stopping ORACLE holds error at the nominal level uniformly over the stream while a naive repeated-looks baseline inflates toward certainty, and it does so without sacrificing competitive structure recovery, a controllable change-detection trade-off, or streaming throughput.

Several limitations point to future work. Orientation uses the bivariate additive-noise asymmetry; with multiple parents and strong nonlinearity the pairwise residual may be imperfect, and a fully parent-conditioned orientation test would strengthen identifiability. The guarantees rest on each increment being a valid e-value under additive structure; when this is violated ORACLE surfaces the conflict by leaving edges undirected rather than committing, which is honest but reduces orientation recall. Scaling to large p is bounded by the quadratic number of candidate pairs; sketching or Markov-blanket-restricted candidate generation would extend the reach. Finally, the same betting machinery applies directly to interventional and time-lagged settings, which we leave for future study.

Table 12: Ablation: witness standardisation (SKIT-level power study on a fixed dependent stream). Standardisation turns the small HSIC signal into an order-one payoff.

standardisation	rejection rate	median detection
on	1.00	84
off	1.00	399

Code and Data Availability

A complete reference implementation of ORACLE, together with the synthetic data generators, the full evaluation suite, and scripts that reproduce every figure and table in this paper, is openly available at <https://github.com/vinhqdang/anytime-causal>. All experiments use synthetic data generated by the released code from fixed random seeds; because the method is a streaming, stochastic procedure, re-runs may differ negligibly from the reported values, and the repository regenerates results, figures, and tables end-to-end.

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